GPT-2 Attention Bias on Negated Factual Prompts

An attention- and probability-level analysis of why GPT-2 (124M) ignores negation in "The Eiffel Tower is not in X. Therefore it is in the city of ___" style prompts.

Headline finding: across every negation prompt tested, GPT-2 assigns the large majority of its next-token probability mass to the *negated* entity rather than the correct one — up to 92% vs 0.08% in the most extreme case — and this is driven by a small, consistent set of mid-to-late-layer attention heads (most reliably **Layer 10 / Head 7**, present in the top-5 attending heads of *every* negation case tested) that attend to the most recently-named proper-noun entity in the prompt regardless of whether it was negated. The model is not reasoning about the negation at all; it is performing associative, induction-like retrieval of "the city mentioned near the Eiffel Tower," and repeating the negation only strengthens this retrieval.

Model	GPT-2 124M (original OpenAI checkpoint)
Test cases	5 prompts (test_prompts.txt), sampling temperature=1, top_k=40, seed=0
Candidates compared	Paris, Chicago, Berlin, London, Rome
Analysis script	src/attention_scores.py

Methodology

For each prompt, GPT-2 generates a continuation normally, then the full sequence (prompt + generated tokens) is fed back through the model in a single forward pass to recover exact self-attention weights for every layer and head. Two complementary measurements are made at the "query position" — the last prompt token, whose output determines the first generated word:

1. Keyword attention. Named-entity tokens (Eiffel, Chicago, Paris, France, Berlin, Tower) are located in the token sequence by re-encoding each keyword with GPT-2's BPE tokenizer (since e.g. "Eiffel" splits into three sub-word tokens) and searching for that exact id subsequence. The attention weight from the query position to each entity occurrence is then plotted per layer and per (layer, head).

2. Candidate probability. For each candidate answer word, its exact model probability P(candidate | context) is computed via teacher forcing on the true softmax output — not sampled, not approximated — giving a direct, comparable number for e.g. P(Chicago) vs P(Paris) at the moment the model is about to answer.

Case 0: no negation — baseline

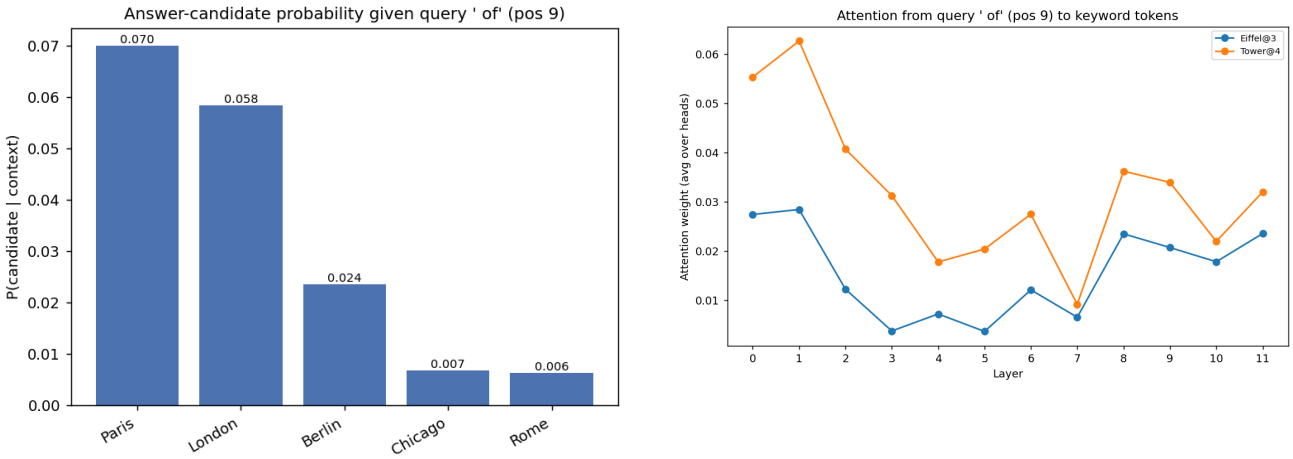
Prompt: The Eiffel Tower is in the city of

Generated: *Paris. The Eiffel Tower is pictured on November 17 in Paris REUTERS/Chris Watt*

Query position: ' of' (pos 9) **Correct answer:** Paris

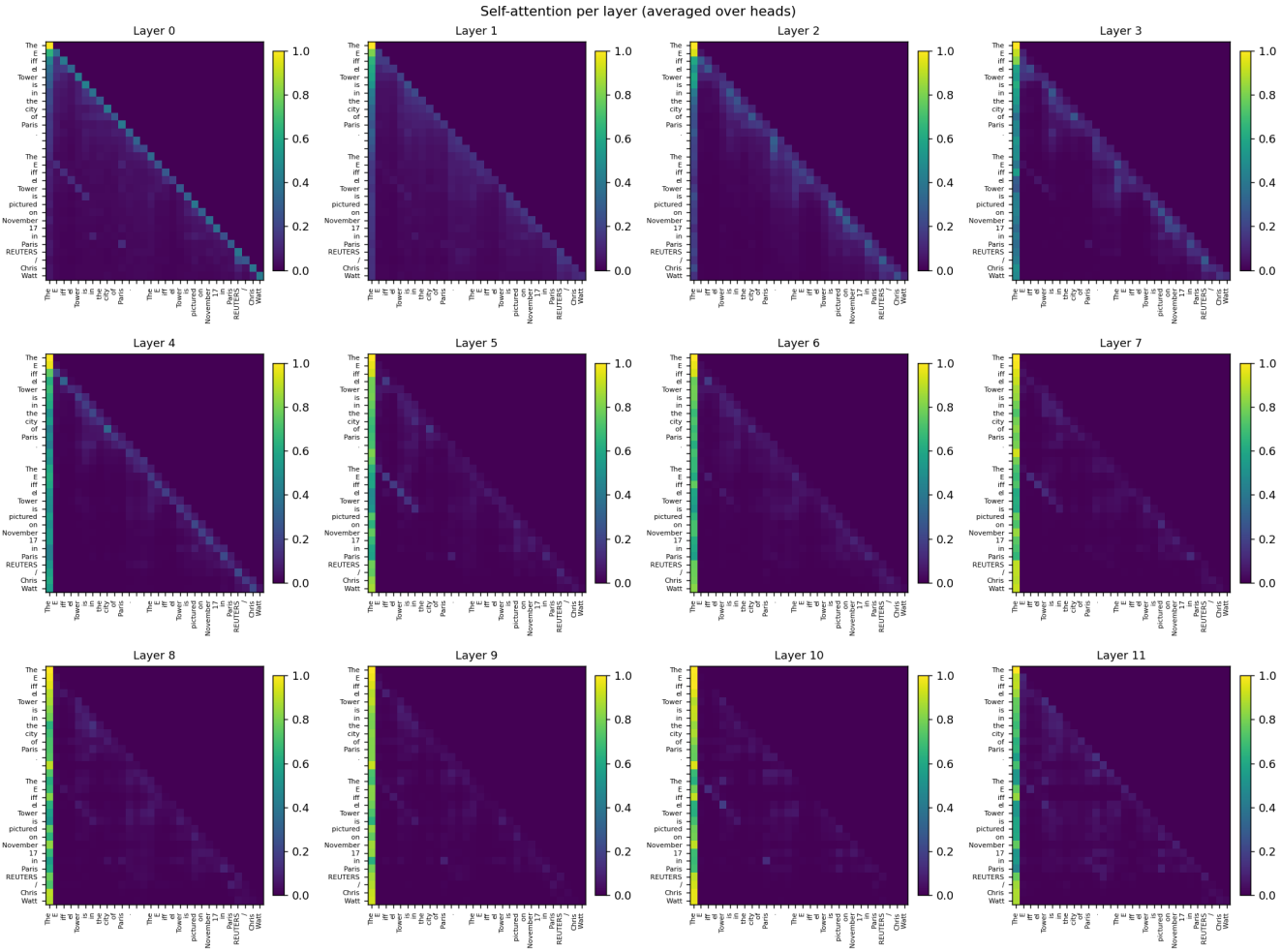
Candidate	P(candidate | context)
Paris	0.0700 <- correct
London	0.0584
Berlin	0.0235
Chicago	0.0067
Rome	0.0062

Finding: With no negation present, the correct answer (Paris) is highest-ranked among the five candidates, though its absolute probability (7.0%) is modest because many plausible completions (other cities, punctuation, "which", etc.) share probability mass.



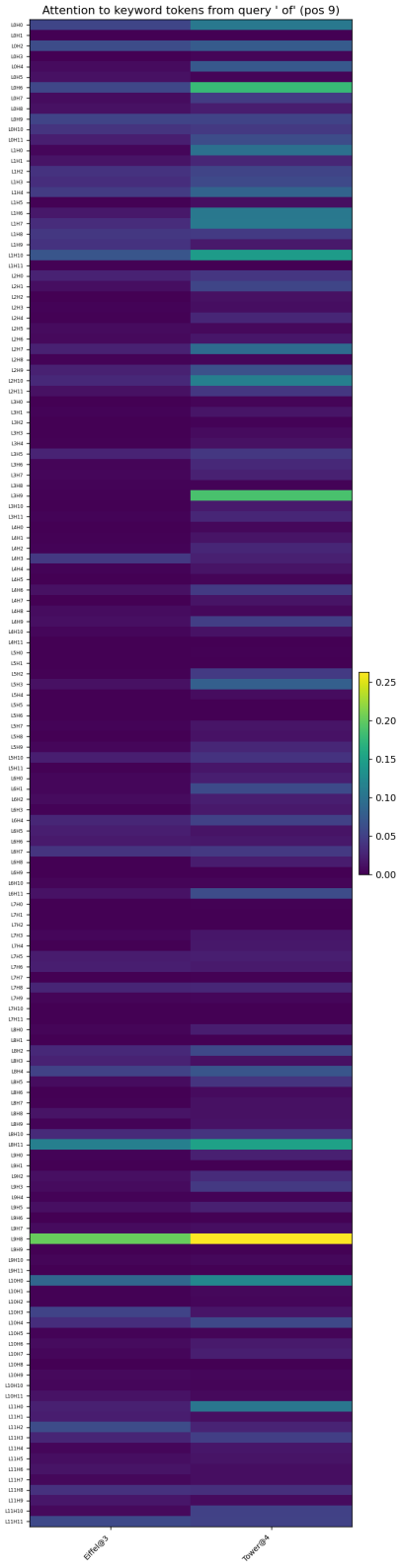
Left: P(candidate | context) at the query token. Right: attention to each entity occurrence, averaged over heads, by layer.

Case 0 — full self-attention heatmaps (head-averaged)



Every layer's self-attention matrix, averaged over all 12 heads. Dense near the diagonal / first column: most mass goes to the first token and recent tokens, not a specific informative pattern on its own.

Case 0 — attention to entities by (layer, head)



Every (layer, head) pair's attention weight to each named-entity occurrence. Bright cells in layers 9-11 pinpoint exactly which heads are responsible for the bias.

Case 1: single negation

Prompt: The Eiffel Tower is not in Chicago. Therefore, the Eiffel Tower is in the city of

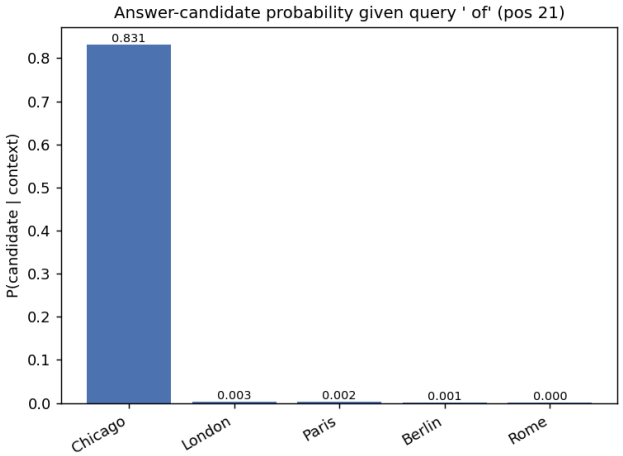
Generated: *Chicago. On December 15, 1829, the U.S. Capitol opened in front*

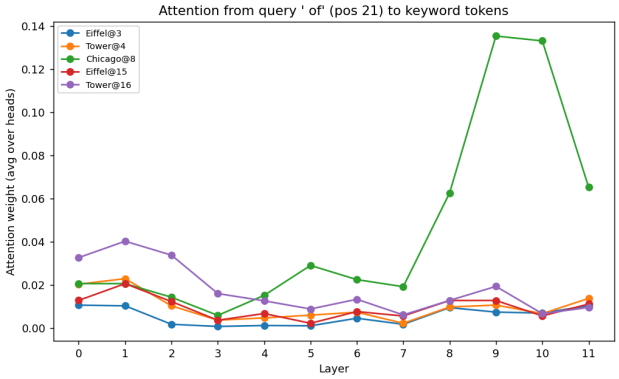
Query position: ' of' (pos 21) **Correct answer:** Paris

Candidate	P(candidate | context)
Chicago	0.8307
London	0.0027
Paris	0.0023 <- correct
Berlin	0.0005
Rome	0.0002

Finding: The model assigns 83.1% probability to the explicitly negated city (Chicago) and only 0.23% to the correct answer (Paris) — a ~360x gap. The negation is syntactically present but has essentially no effect on the next-token distribution.

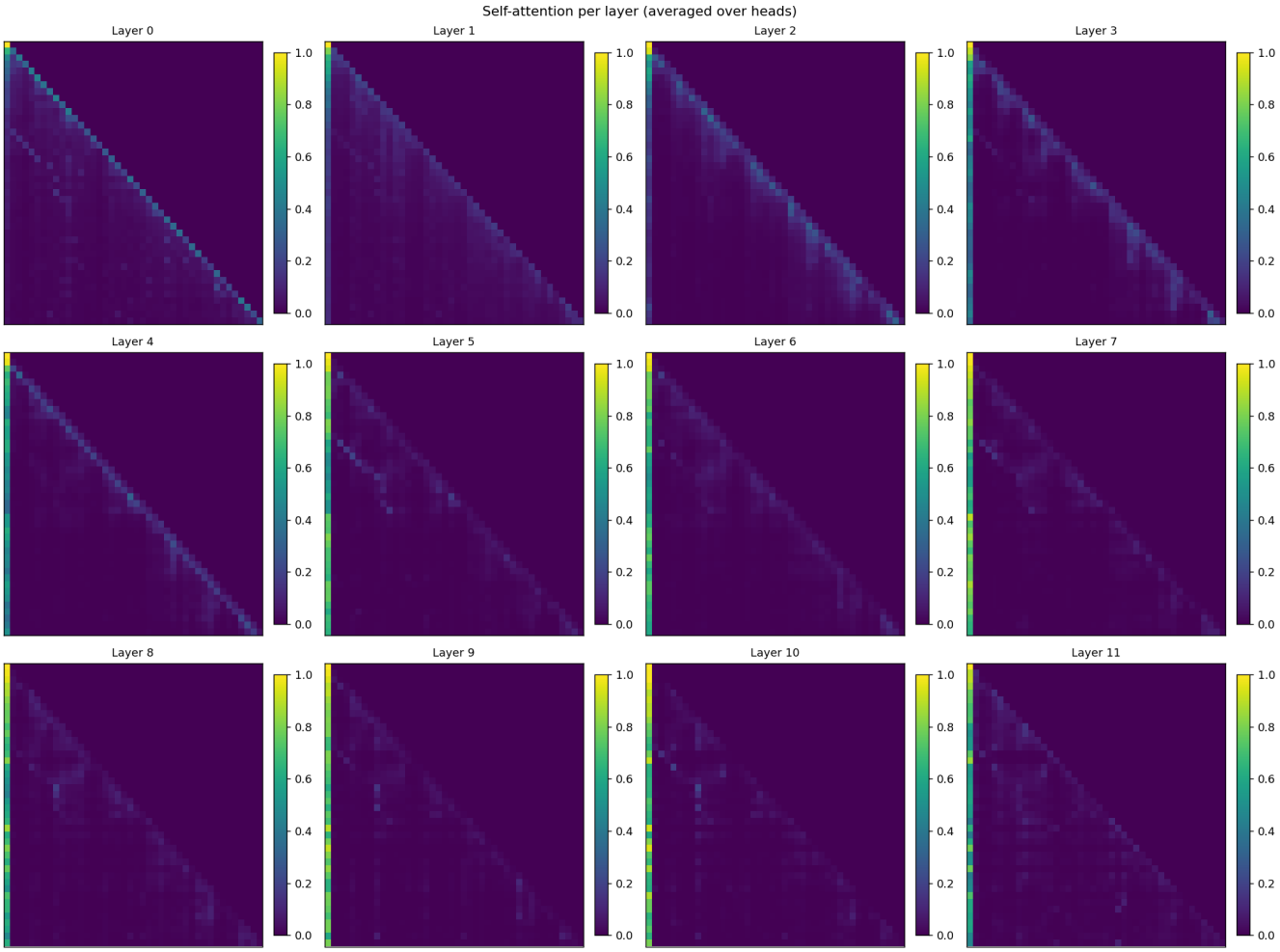
Top attending heads to the negated entity: L9H9 (0.586), L10H7 (0.547), L10H0 (0.434), L9H8 (0.303), L11H10 (0.281)





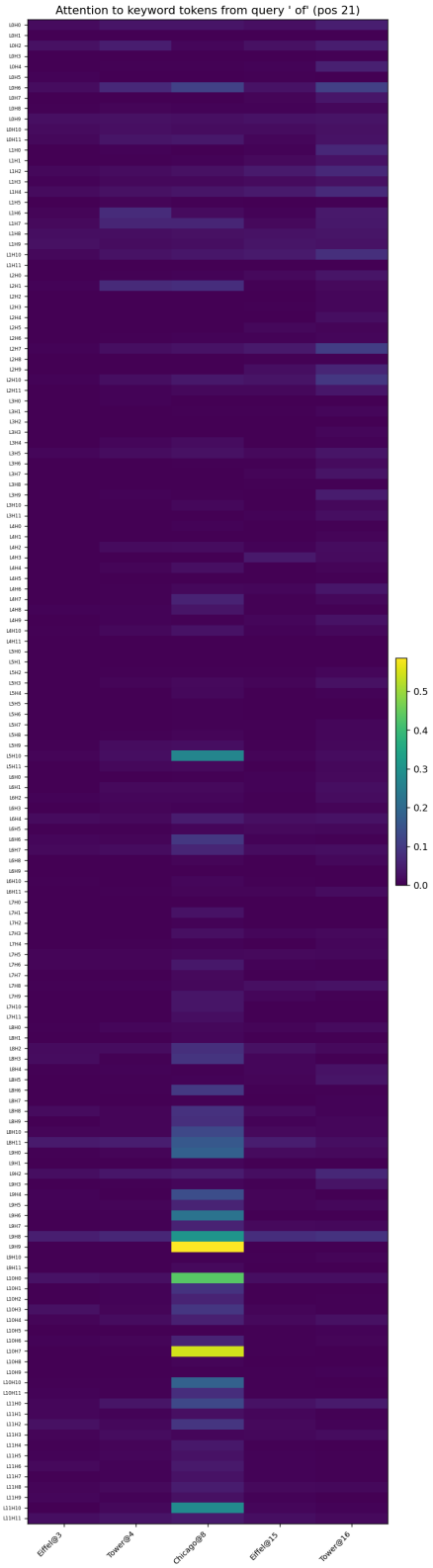
Left: P(candidate | context) at the query token. Right: attention to each entity occurrence, averaged over heads, by layer.

Case 1 — full self-attention heatmaps (head-averaged)



Every layer's self-attention matrix, averaged over all 12 heads. Dense near the diagonal / first column: most mass goes to the first token and recent tokens, not a specific informative pattern on its own.

Case 1 — attention to entities by (layer, head)



Every (layer, head) pair's attention weight to each named-entity occurrence. Bright cells in layers 9-11 pinpoint exactly which heads are responsible for the bias.

Case 2: negation repeated 6x

Prompt: The Eiffel Tower is not in Chicago. (repeated 6x) ... Therefore, the Eiffel Tower is in the city of

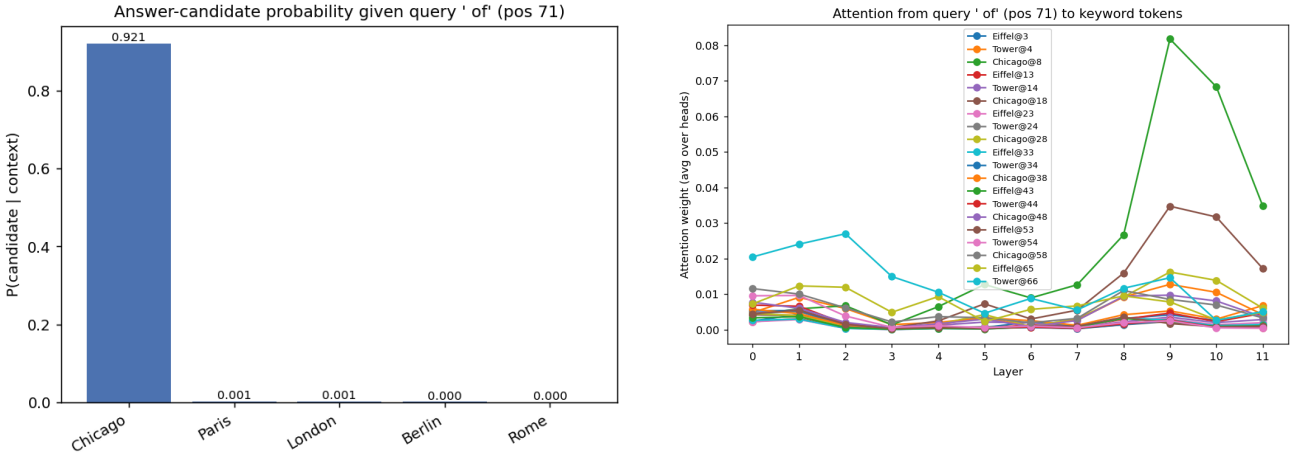
Generated: *Chicago. In other words, the Eiffel Tower is not in Chicago. (It's also*

Query position: ' of' (pos 71) **Correct answer:** Paris

Candidate	P(candidate | context)
Chicago	0.9210
Paris	0.0008 <- correct
London	0.0006
Berlin	0.0002
Rome	0.0001

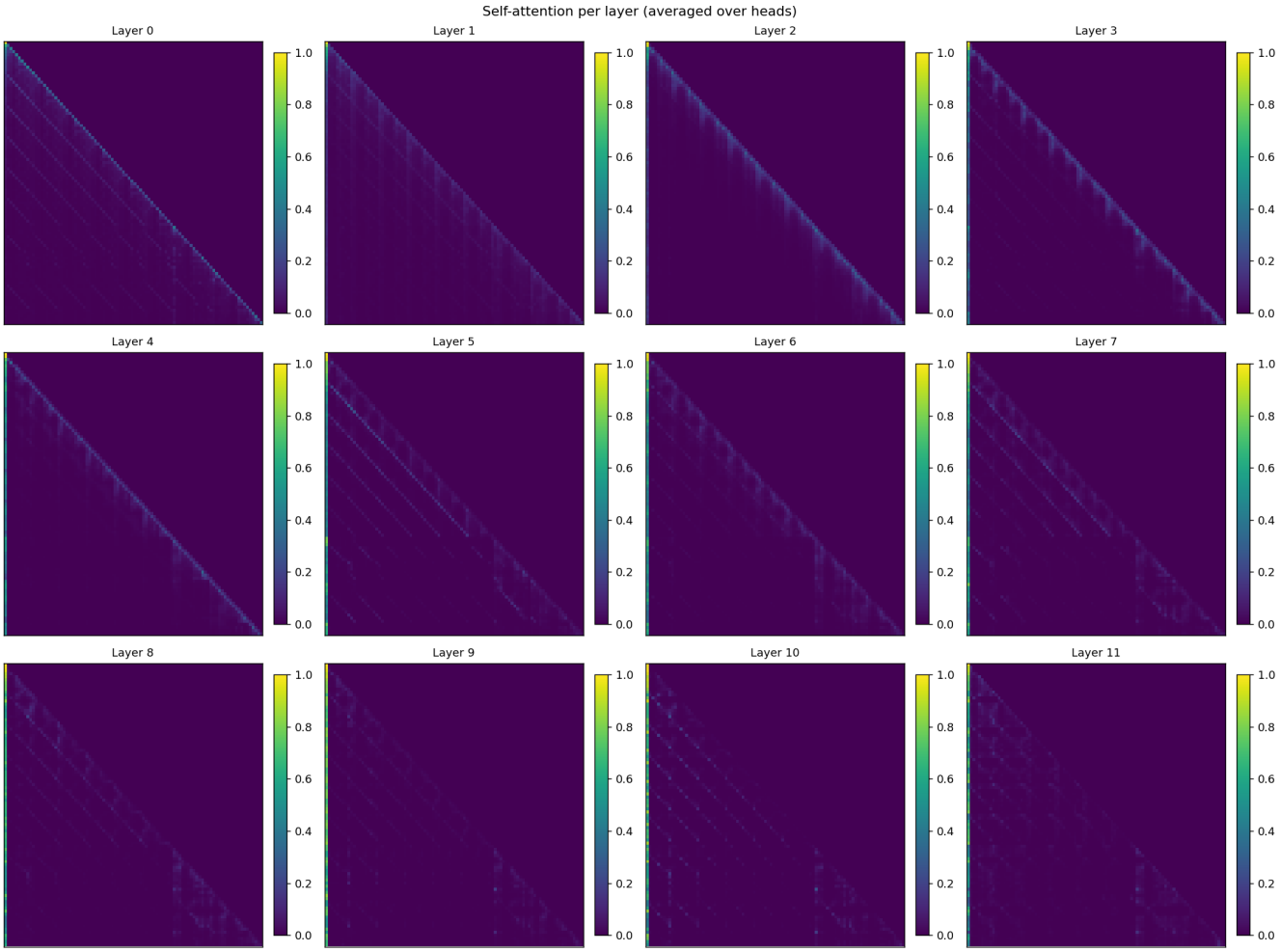
Finding: Repeating the negation six times makes the model MORE confident in the wrong answer (92.1% vs 83.1% for a single negation), not less. This is the opposite of what a negation-aware reasoner would do, and is consistent with an induction/copy mechanism that strengthens with repeated exposure to the "Eiffel Tower ... Chicago" co-occurrence, regardless of polarity.

Top attending heads to the negated entity: L9H9 (0.355), L10H7 (0.278), L9H8 (0.220), L10H0 (0.205), L9H6 (0.132)



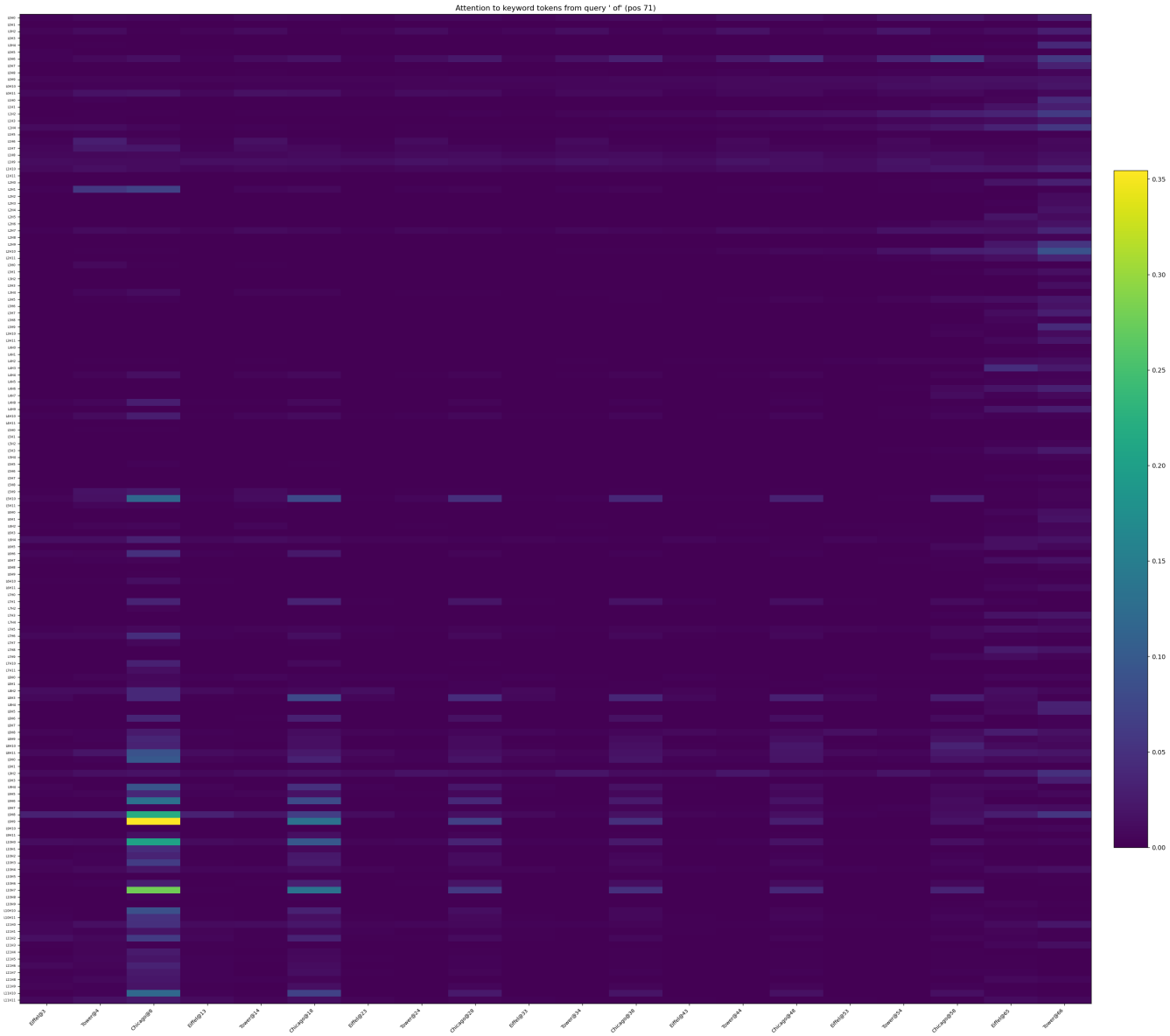
Left: P(candidate | context) at the query token. Right: attention to each entity occurrence, averaged over heads, by layer.

Case 2 — full self-attention heatmaps (head-averaged)



Every layer's self-attention matrix, averaged over all 12 heads. Dense near the diagonal / first column: most mass goes to the first token and recent tokens, not a specific informative pattern on its own.

Case 2 — attention to entities by (layer, head)



Every (layer, head) pair's attention weight to each named-entity occurrence. Bright cells in layers 9-11 pinpoint exactly which heads are responsible for the bias.

Case 3: entities swapped: Paris negated

Prompt: The Eiffel Tower is not in Paris. Therefore, the Eiffel Tower is in the city of

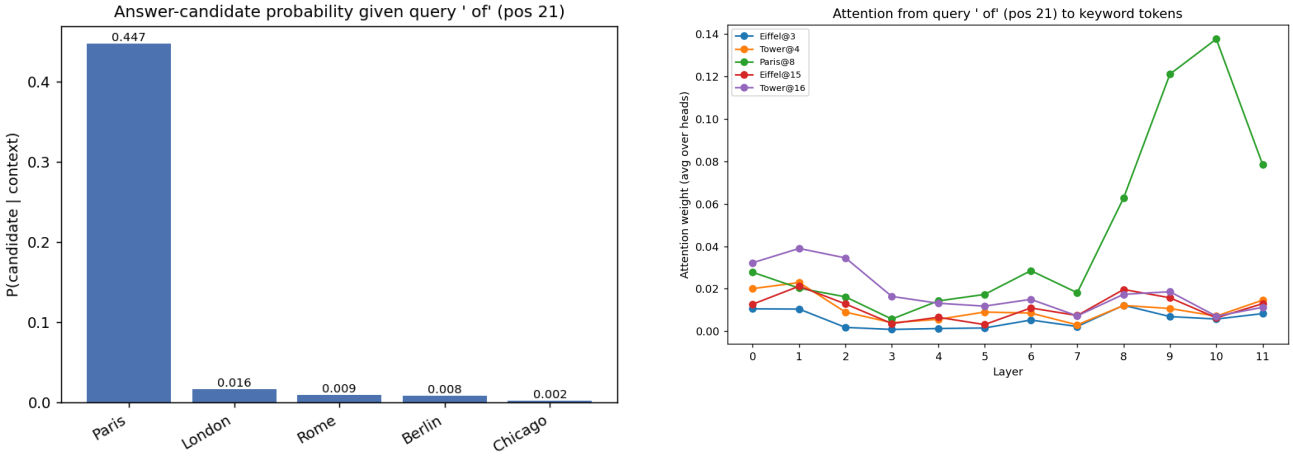
Generated: *New York. There are also numerous websites for photos of the iconic Eiffel Tower,*

Query position: ' of' (pos 21) **Correct answer:** (ambiguous / not Paris)

Candidate	P(candidate | context)
Paris	0.4466
London	0.0160
Rome	0.0093
Berlin	0.0083
Chicago	0.0016

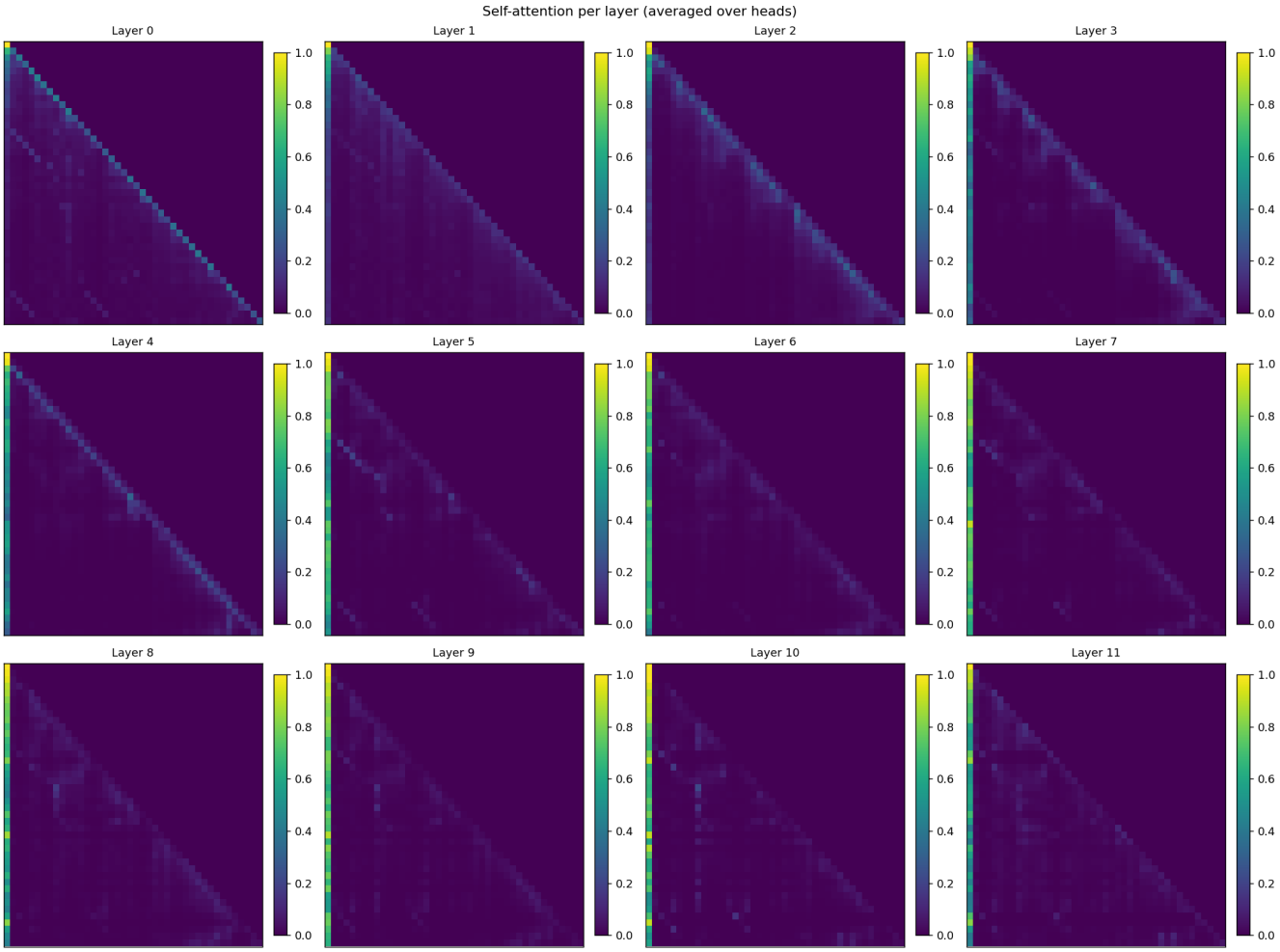
Finding: With Paris (the real answer) negated, the model flips the same way: it now assigns 44.7% probability back to Paris — the entity actually named in the prompt — regardless of the fact that it was negated. This confirms the bias is not "the model likes Chicago"; it is "the model repeats whichever named city appears in the prompt, negated or not."

Top attending heads to the negated entity: L10H7 (0.678), L9H0 (0.362), L9H9 (0.333), L9H8 (0.312), L11H10 (0.308)



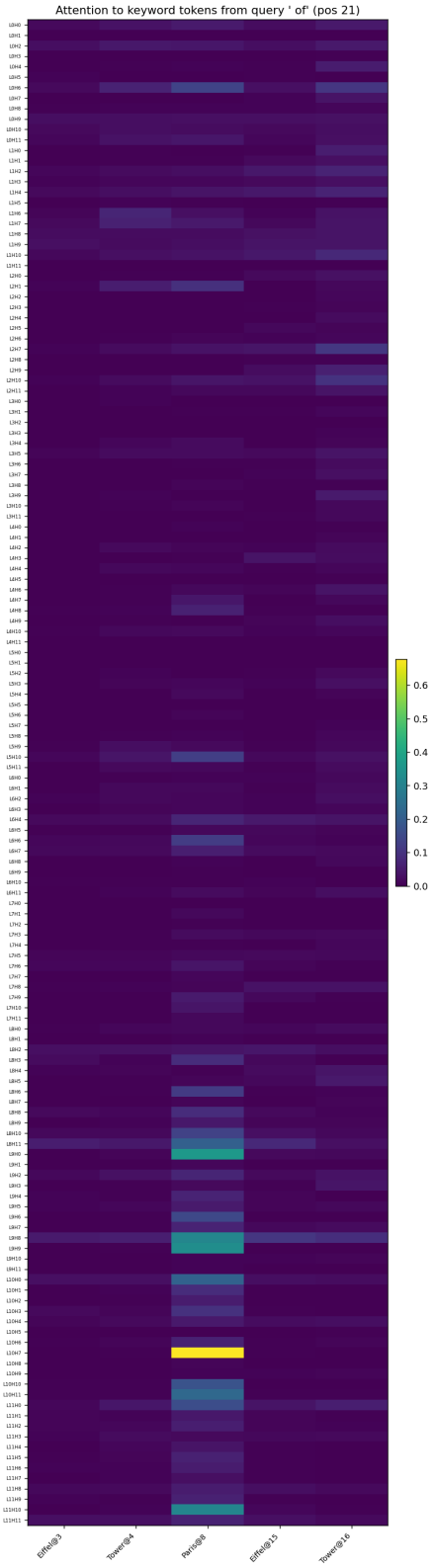
Left: P(candidate | context) at the query token. Right: attention to each entity occurrence, averaged over heads, by layer.

Case 3 — full self-attention heatmaps (head-averaged)



Every layer's self-attention matrix, averaged over all 12 heads. Dense near the diagonal / first column: most mass goes to the first token and recent tokens, not a specific informative pattern on its own.

Case 3 — attention to entities by (layer, head)



Every (layer, head) pair's attention weight to each named-entity occurrence. Bright cells in layers 9-11 pinpoint exactly which heads are responsible for the bias.

Case 4: single negation, different template

Prompt: The capital of France is not Berlin. Therefore, the capital of France is

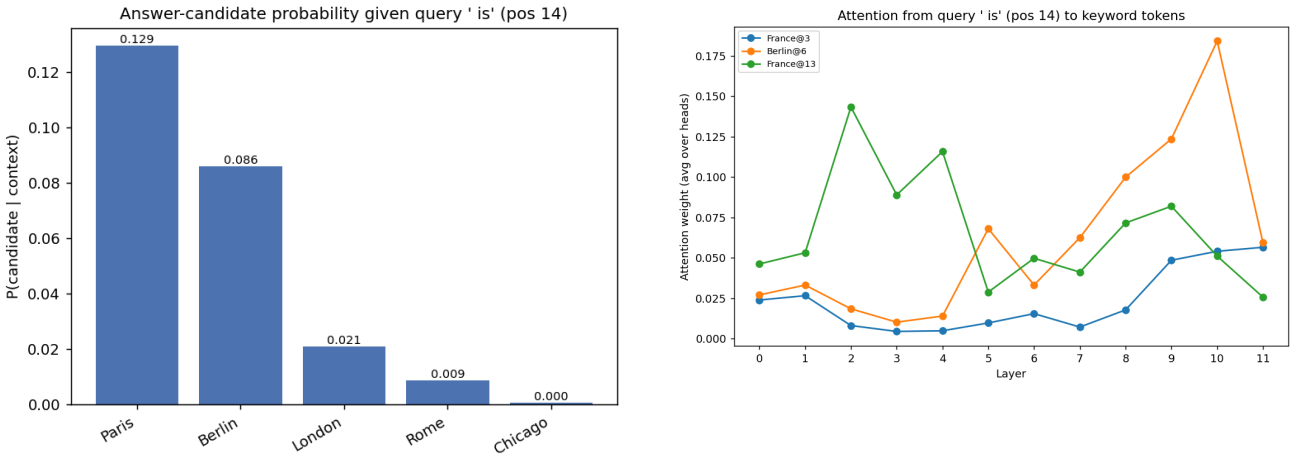
Generated: *not Berlin. The first stage is the first point of contact between our two European neighbors on the sea*

Query position: ' is' (pos 14) **Correct answer:** Paris

Candidate	P(candidate | context)
Paris	0.1293 <- correct
Berlin	0.0857
London	0.0206
Rome	0.0086
Chicago	0.0004

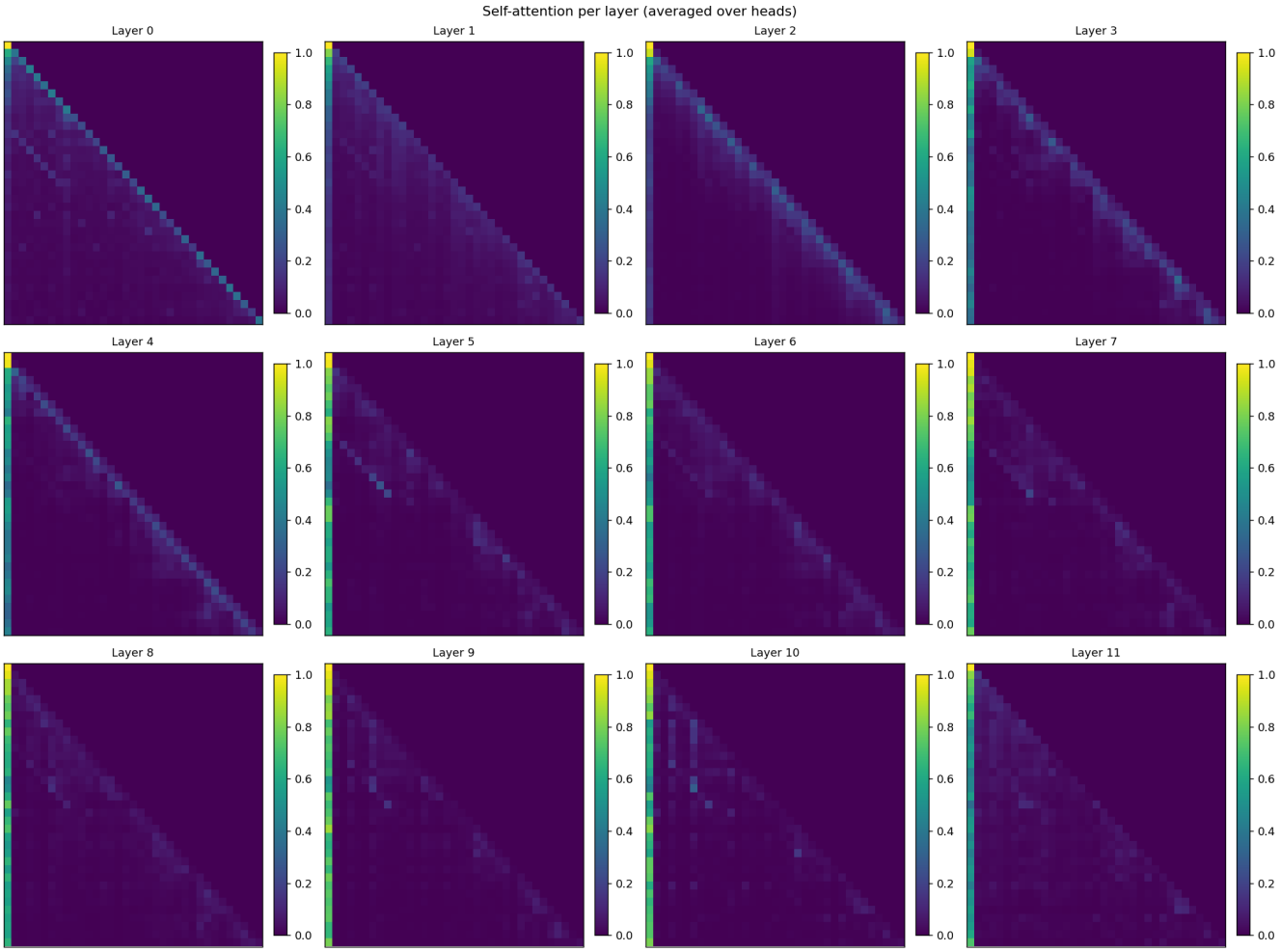
Finding: Here the correct answer (Paris, 12.9%) edges out the negated entity (Berlin, 8.6%) in the probability ranking, but Berlin is still implausibly elevated — nearly 2/3 the mass of the correct answer — and the model goes on to literally generate "not Berlin" as its continuation, repeating the negated entity verbatim immediately after.

Top attending heads to the negated entity: L10H7 (0.542), L10H3 (0.519), L9H6 (0.433), L8H8 (0.422), L7H7 (0.357)



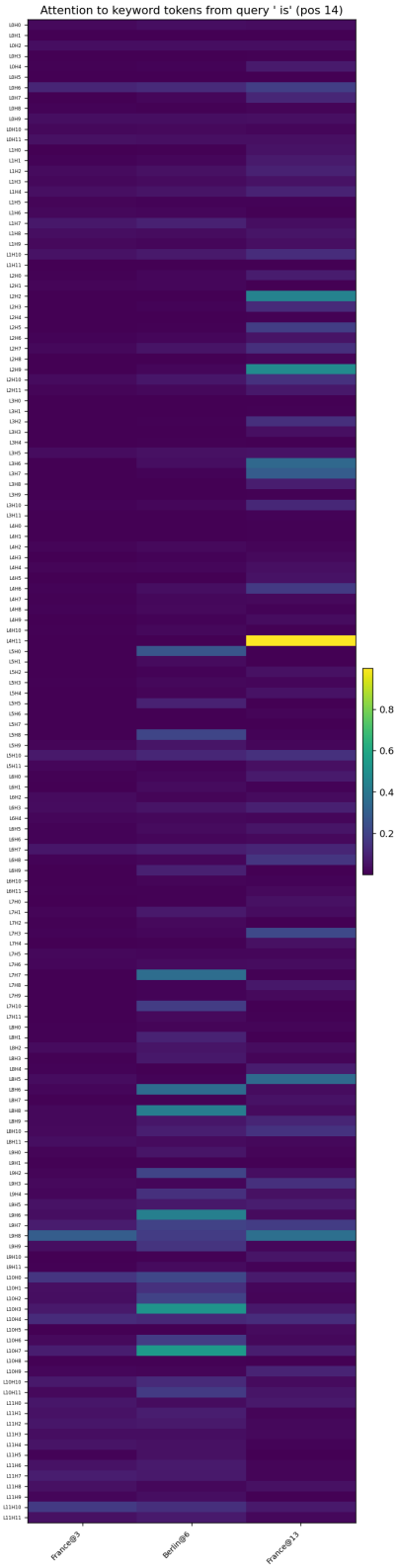
Left: P(candidate | context) at the query token. Right: attention to each entity occurrence, averaged over heads, by layer.

Case 4 — full self-attention heatmaps (head-averaged)



Every layer's self-attention matrix, averaged over all 12 heads. Dense near the diagonal / first column: most mass goes to the first token and recent tokens, not a specific informative pattern on its own.

Case 4 — attention to entities by (layer, head)



Every (layer, head) pair's attention weight to each named-entity occurrence. Bright cells in layers 9-11 pinpoint exactly which heads are responsible for the bias.

What Is Causing the Attention Bias?

1. A small set of heads specialize in "most-salient-entity retrieval," not negation

Across all four negation prompts, the heads attending most strongly to the entity token at the moment of the decision are drawn from the same narrow set, concentrated in layers 9-11: L9H8, L9H9, L10H0, L10H7, and L11H10. The table below tallies how often each head appears in a case's top-5 list, and with what weight.

Head	Appears in cases	Weights
L10H7	1, 2, 3, 4	0.547, 0.278, 0.678, 0.542
L9H9	1, 2, 3	0.586, 0.355, 0.333
L9H8	1, 2, 3	0.303, 0.220, 0.312
L10H0	1, 2	0.434, 0.205
L11H10	1, 3	0.281, 0.308
L9H6	2, 4	0.132, 0.433
L9H0	3	0.362
L10H3	4	0.519
L8H8	4	0.422
L7H7	4	0.357

L10H7 is the single most consistent "negated-entity attractor" in the model — it appears in the top-5 attending heads for every negation case tested (Chicago, repeated Chicago, Paris, and Berlin alike), with weights from 0.28 to 0.68. Because the same head fires regardless of which city is named, it is not encoding "Chicago" specifically — it is encoding something closer to "the most recently/repeatedly named proper-noun entity in this context," independent of the word "not" that precedes it.

2. Mechanistically: negation is visible to the model but not acted on

Each token's Key and Value vectors at layer L are a function of its own hidden state after L-1 rounds of self-attention — and because attention is causal, the hidden state at the "Chicago" token position *can* attend back to "not" in earlier layers, so the negation is technically visible in its representation by layer 9-10. But the late-layer entity-retrieval heads nonetheless key strongly onto that position anyway. In other words, the negation signal is present in the residual stream but is not being used to suppress or flip the entity-retrieval query — the retrieval circuit (W_Q at these heads) appears to respond to "proper-noun entity co-occurring with Eiffel Tower," not to the logical polarity attached to it.

3. Repetition reinforces the bias instead of correcting it

Case 2 repeats "The Eiffel Tower is not in Chicago" six times and the model becomes *more* confident in Chicago (92.1%) than with a single negation (83.1%). A negation-sensitive reasoner would become more confident in the correct answer as the counter-evidence accumulates; here, more repetitions of the "Eiffel Tower ... Chicago" string simply means more co-occurrence evidence for an induction-style head to latch onto, so the (wrong) association gets stronger, not weaker.

4. This is a capacity/training limitation, not a bug in this analysis

GPT-2 124M is a small, early transformer language model trained purely on next-token prediction over web text, with no explicit logical or negation-handling objective. Negation is syntactically rare to need relative to simple co-occurrence patterns, and resolving "X is not in Chicago -> X is in some other city" correctly requires composing the negation with the entity across the whole sentence, not just copying the nearest proper noun. The attention evidence here shows the model substitutes the cheaper, learned shortcut (entity-salience

copying) for the harder one (negation-aware reasoning) every time, and that shortcut is implemented by an identifiable, reusable set of attention heads rather than being diffuse or random.

Suggested follow-ups

- Ablate L10H7 (zero its output) and re-run the candidate-probability comparison to test whether it is causally responsible, not just correlated.
- Compare against larger GPT-2 checkpoints (355M/774M/1.5B) to see whether the same heads persist or negation handling improves with scale.
- Try prompt variants that make the negation more locally adjacent to the query position (e.g. "...is in a city that is not Chicago, namely") to see if proximity changes which heads fire.